

Results of a comprehensive monitoring program at Pratte's Reef

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Introduction

Pratte's Reef, located in El Segundo California, is the first structure constructed in United States waters that was entirely conceived, designed and built as an enhancement to recreational surfing with the intent of mitigating the loss of surfing areas due to the construction of the El Segundo Groin and associated backfilling on Dockweiler Beach. Pratte's reef is named after Tom Pratte, an environmental activist who spearheaded the effort hold coastal developers responsible for the destruction or damage of surfing areas.

Site Description

Pratte's Reef is located offshore of Dockweiler Beach near the center of Santa Monica Bay in El Segundo, California. Specifically, the reef is approximately 200 m south of the Hyperion Sewage Treatment Plant's 1-mile outfall. This is a well-known surfing spot referred to as "Shitpipe". Access to the reef site is through the newly constructed parking lot located approximately 1 mile south of the intersection of Imperial Highway and Vista Del Mar St. These locations are shown in Figure 1.

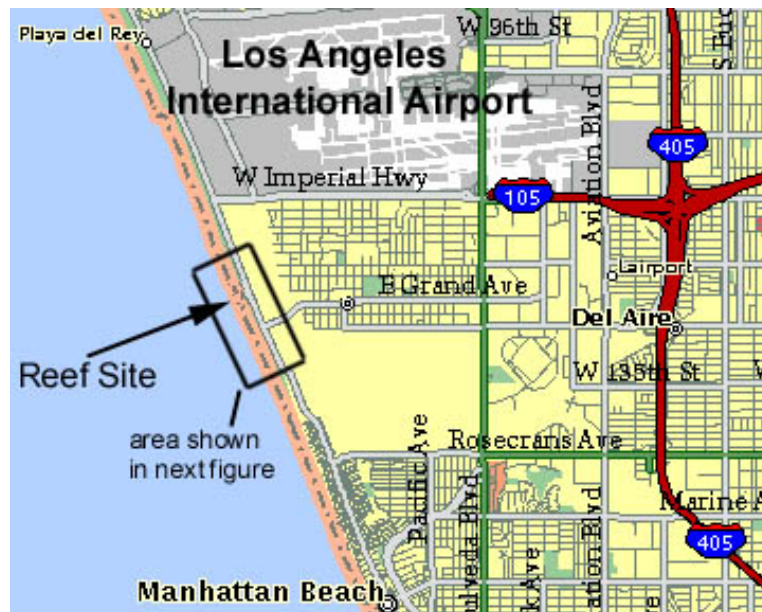


Figure 1: Map showing the location of Pratte's Reef in El Segundo, CA. El Segundo is located 15 km west of downtown Los Angeles.

This portion of Dockweiler Beach is a wide sandy beach, typical of the west-facing coastline of Santa Monica Bay. The reef site lies along a heavily armored shoreline and is within 500 m of two coastal structures, the Hyperion 1-mile Outfall to the north and the Grand Street Jetty to the south.

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The El Segundo Jetty shown in Figure 2 is the southernmost of a several shore perpendicular structures that affect the surf zone in the 5-mile stretch of beach between Playa Del Rey and El Segundo.



Figure 2: A section of an aerial photograph showing the location of the reef relative to other coastal structures. The black lines crossing the beach correspond to range lines used in shoreline profile monitoring. The surfing beach known as El Porto is out of frame to the right, approximately 1 km south of the El Segundo Jetty.

The beach fronting the reef site is approximately 60 to 120 meters wide depending on the tide and the season. It is partially backed by a ridge of low sand dunes approximately 8 m high. The beach slopes gently from the base of the dunes until the back beach area meets the shore face, which has a much steeper slope.

Design and Construction

Pratte's Reef is a structure designed to modify wave-breaking characteristics to be more applicable for recreational surfboard surfing. Initially, the 'reef' was designed as a small, submerged breakwater composed of 110 sand filled geotextile bags. The bags measure 4' x 7' x 10' and contain a maximum volume of 280 ft³ (7.9 m³). Each bag was filled to between 80 and 90% capacity due to restrictions imposed by the equipment used to fill and close the bags. Each bag weighed approximately 14 tons when filled with sand. On September 22nd, 2000, the bags were filled with material in the Port of Los Angeles, and then loaded onto a barge, which traveled to the site. The bags were placed into the surf zone by barge-mounted crane (Figure 3). The bags are arranged in an obtuse 'V' shape with the apex pointing offshore. The purpose of this arrangement was to force incoming swells to break further from shore than they would normally. The resulting breaker would then have a larger peel angle and be more suitable for surfing. After the initial installation, the depth at the outermost point of the reef was approximately - 6 ft MLLW (mean lower low water, the level to which tide charts are calibrated). Figure 4 shows the design placement of the reef bags.

A second installation of reef bags was performed on April 23rd and 24th 2001. Ninety new bags were installed that increased the reef volume by 80%. The crest width of the reef was widened with the new bags and it was made shallower, to within 3ft of MLLW. The new bags were placed directly over the phase 1 bags, which were largely covered in sand that had moved offshore during the winter.



Figure 3: Lifting the bags off of the barge and placing them in the desired location.

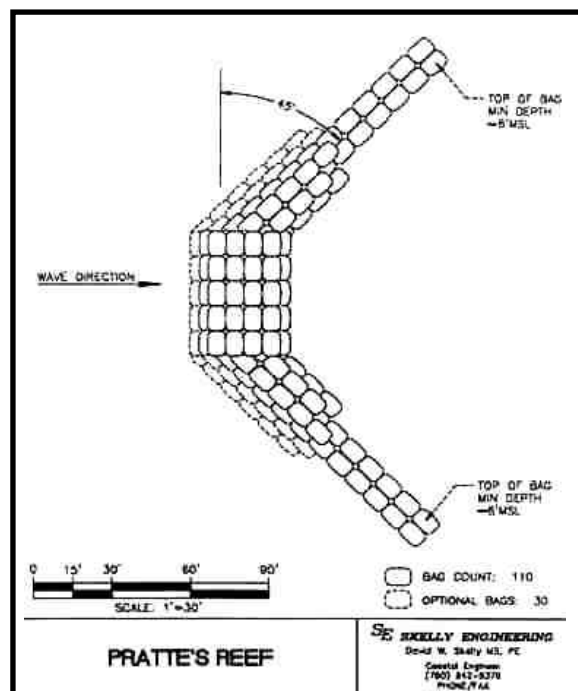


Figure 4: The design layout of Pratte's Reef.

Monitoring Program Methodology

As required by the permits for the construction of Pratte's Reef, a detailed monitoring and reporting program was implemented to quantify the effects of Pratte's Reef on local surfing conditions, shoreline features and near shore bathymetry. The survey also qualitatively described the condition of the reef bags and effects on the local biology. A monitoring plan was outlined during the design and construction phases of the reef project to accomplish these goals. A limited monitoring program was conducted prior to construction. As required by the permits, the entire monitoring program was implemented immediately after construction. The monitoring plan consisted of four parts: shoreline profiling, bathymetric surveys, dive surveys and surf quality observations. Each of these is described below.

Shoreline Monitoring – Shoreline monitoring was accomplished through surveying the shoreline profile and photography and the beach and near shore zone. Quantitative measurements are taken using standard rod and level surveying techniques. That is, measuring distances and differences in elevation along fixed range lines that cross the beach from the base of the dune out to wading depth. A total of 6 range lines were used in this project. Their locations relative to the reef are shown in Figure 2. The distance from the northern line to the southern line is approximately 550 m.

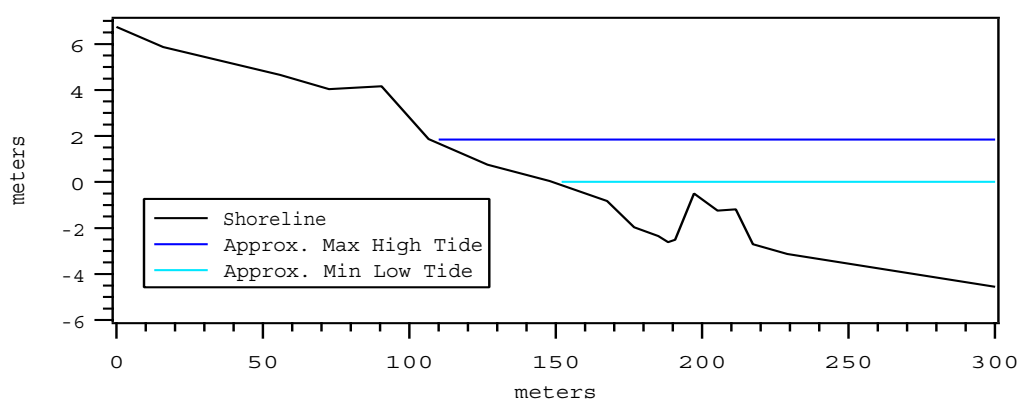


Figure 5: A sample cross-shore profile take along the centerline of Pratte's Reef in October 2001.

Bathymetric Surveys – The most complex part of the monitoring program was the gathering offshore depth data. This required the use of accurate depth and position recording instruments. The instruments were mounted on a small boat, which was piloted along lines perpendicular to the shore. An Acoustic Doppler Current Profiler (ADCP) recorded depths and current data. Absolute position information was recorded using a differential global positioning system (DGPS). Both streams of data were recorded simultaneously by an on board laptop computer. The position information is presented in meters in the Universal Transverse Mercator (UTM) coordinate system. Bathymetric surveys were performed quarterly.

Dive Surveys – Dive surveys were used to assess the condition of the reef bags and the level of biological growth associated with the reef. An underwater video camera and handheld depth gauge

were used to record the condition of the bags and get precise depth information. Divers took notes and assessed bag locations and patterns of sediment deposition or scour. Dive surveys were difficult to accomplish due mainly to poor visibility.

Surf Quality – Volunteers were used to take quasi-daily reports of surf conditions at the reef site and at various locations along nearby beaches. Surf quality monitoring was conducted by having the volunteer fill out a standardized Surf Environment Observation (SEO) form. The SEO form was based on the U.S. Army Corps of Engineer's Littoral Environment Observation (LEO) form. Categories on this form include estimates of wave height and period, a count of the number of surfers present, weather conditions and an estimate of the near shore current direction and strength. The data were entered into a database for analysis and the results are presented below.

Results of the Monitoring Program

Profiles

The beach in the vicinity of the reef has not been affected by the presence of the reef in any quantifiable way. The shoreline morphology has basically adhered to the seasonal changes that one would expect along a beach of this type. That is, a landward recession of the beach berm in response to large waves (winter profile) and a seaward proccession of the beach berm during the time of smaller waves (summer profile). For the beaches in this area, the full winter profile is usually reached after several significant swells in the fall and early winter (October through January). The beaches remained at their winter profile until the spring when they began to widen towards the summer profile. The full summer profile is usually seen in the late summer or early fall (July – September). Figure 6 shows the seasonal trends in the shoreline through year 1 along the center profile. Figure 7, also taken along the centerline of the reef shows how the summer profile has remained very consistent over the 2-year survey period. The November 2000 profile was taken before any significant swells had hit the beach.

The profiling also showed that the high crest of the reef, which was so prominent after the second installation, was gradually eroded away over the subsequent months. Figure 8 shows this progression.

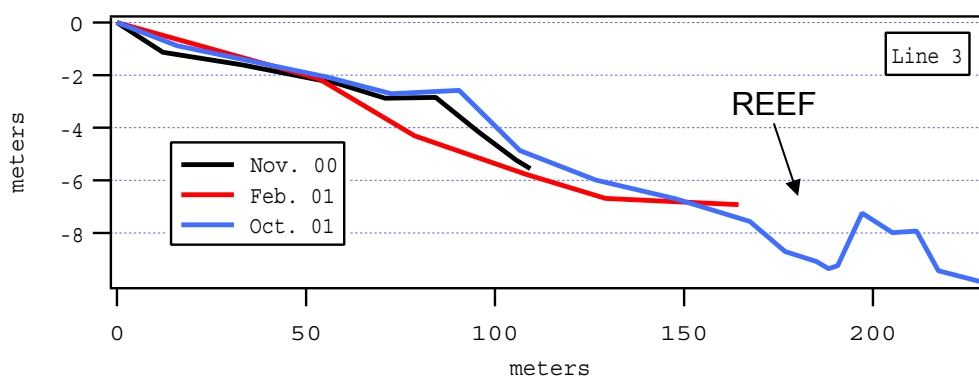


Figure 6: Seasonal Profiles over Year 1 for the reef centerline.

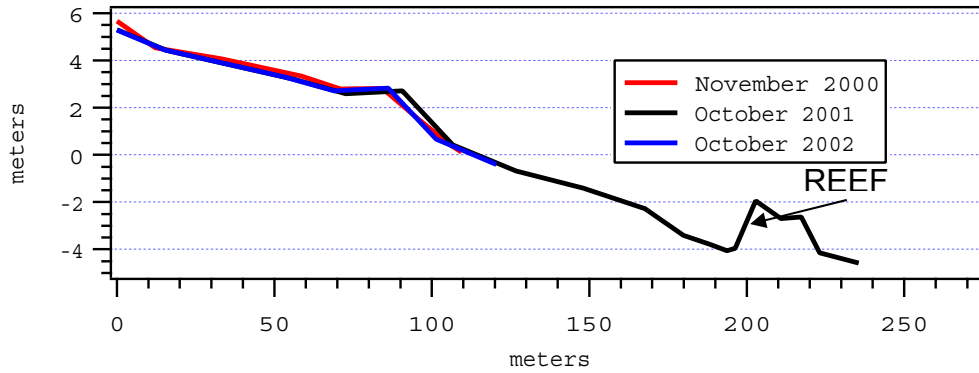


Figure 7: Comparison between three summer profiles.

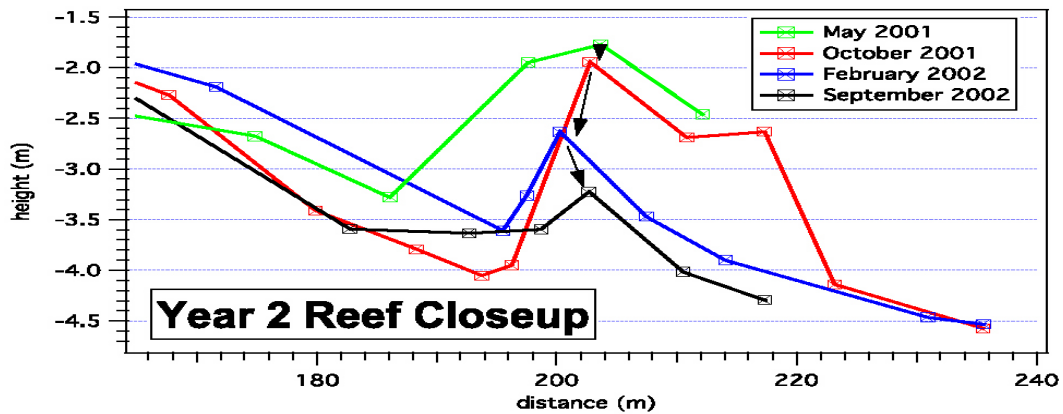


Figure 8: The reef crest eroding after the second installation (April 2001).

Bathymetric Surveys

Pratte's Reef has not had any impact on the offshore bathymetry outside of the contours created by the reef bags themselves. In general, the only noticeable changes in the bathymetric data are due to the natural seasonal changes describe above. During the winter swells, the overall offshore slope is decreased and the contours shift seaward. The trend is reversed in the summer months and the offshore slope becomes steeper. This can be seen in Figure 9, which compares the offshore bathymetry from 2 different seasons.

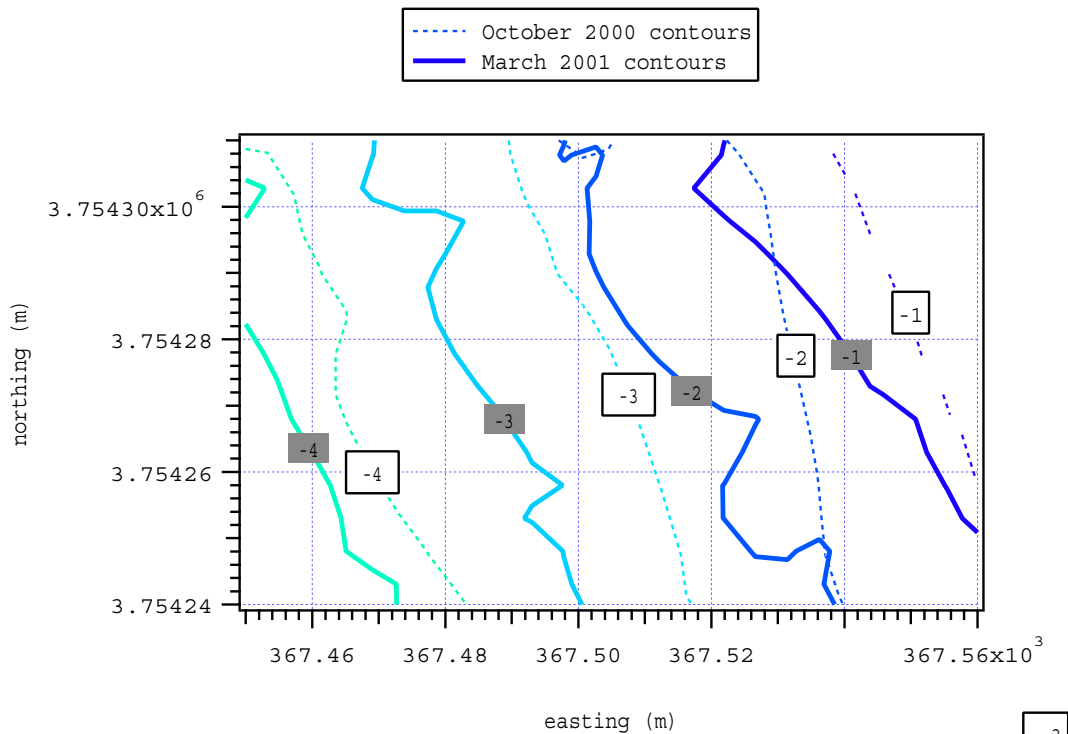


Figure 9: A comparison between bathymetry from a summer profile (October 2000) to a winter profile (March 2001). Note how the contours have moved offshore in response to the large wave that eroded sand off of the beach and deposited it offshore.

Dive Surveys

Dive Surveys were used to both observe the physical condition of the reef and observe the biological conditions on and near the reef. The dive surveys revealed that rapid biological growth took over Pratte's Reef. Within weeks of the initial bag installation, algae had begun to grow on the bags and schools of small fish were attracted to the site. The biological activity increased after the installation of the second set of bags. With more exposed bags, larger crustaceans such as crabs and lobsters colonized the area. The following photographs are from the first and last dive surveys respectively.



Figure 10: A school of small fish over the reef, October 2000, just weeks after the first bags were placed.



Figure 11: Lobsters on the reef in August 2002. Several other species of marine life were also observed, sea stars, halibut, and mussels among others.

The dive surveys also revealed the extent to which reef bags were deteriorating or were covered by sand. Dive surveys conducted in the winter of 2001 showed that the reef was largely buried in sand. After the second installation the reef (and the marine organisms that used it for shelter) were quite visible. A dive survey in August 2002 showed that several reef bag units were ripped or shredded and were losing their fill material. Large torn or shredded sections of the bags were removed upon discovery. These are shown in Figure 12. It should be noted that there were two types of reef bags used, a white bag made of polyester and a black bag made of polypropylene. The weave pattern on

each of the bags was also different. Of the two, the white polyester bags held up much better than the polypropylene. The weave pattern on the white bags did not allow small rips to propagate into big tears as they did on the black material.

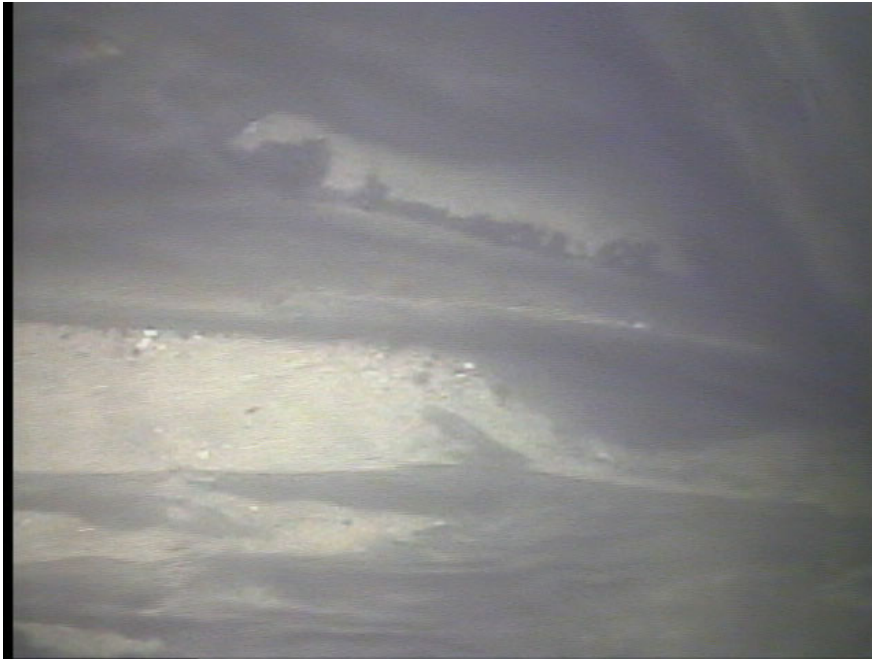


Figure 12: A ripped reef bag in August 2002.

Surf Observations

Volunteers recorded over 427 Surf Environment Observations (SEO) during the two years monitoring of Pratte's reef. Of these, 393 were recorded at the reef site and 34 were recorded at other nearby beaches. The single most important result from these observations is the almost complete lack of surfers at the reef. Although there were a handful of reports of people surfing at the reef, these almost never coincided with an SEO observer being present. This underscores the reality that Pratte's did not perform as well as hoped in terms of surf quality improvement. The breakdown of SEO observations by month is shown in Figure 13.

Figure 14 illustrates the number of surfers present at Pratte's Reef. This plot is somewhat misleading however. There were at least five additional known occasions where people were surfing at Pratte's Reef, but there was no one there to record an official SEO. Even with these additional surfers counted, the total number of people that surfed at Pratte's reef is certainly less than 25 over two years, hardly making it a legitimate surf spot. SEO results reporting on the number of surfers at other locations are misleading. Some of those SEO's were taken at locations where nobody generally surfs due to poor waves quality (i.e. Dockweiler Beach north of the Hyperion 1-mile Outfall) or the SEO was recorded at an unusually early hour (before 6 am!) when even the most hardcore surfers are still checking the waves or putting on their wetsuits. The SEO program could have been improved by ensuring that a more representative sample of alternate locations was recorded. Even a casual observer can see that on an average day the number of surfers on nearby beaches is in the hundreds.

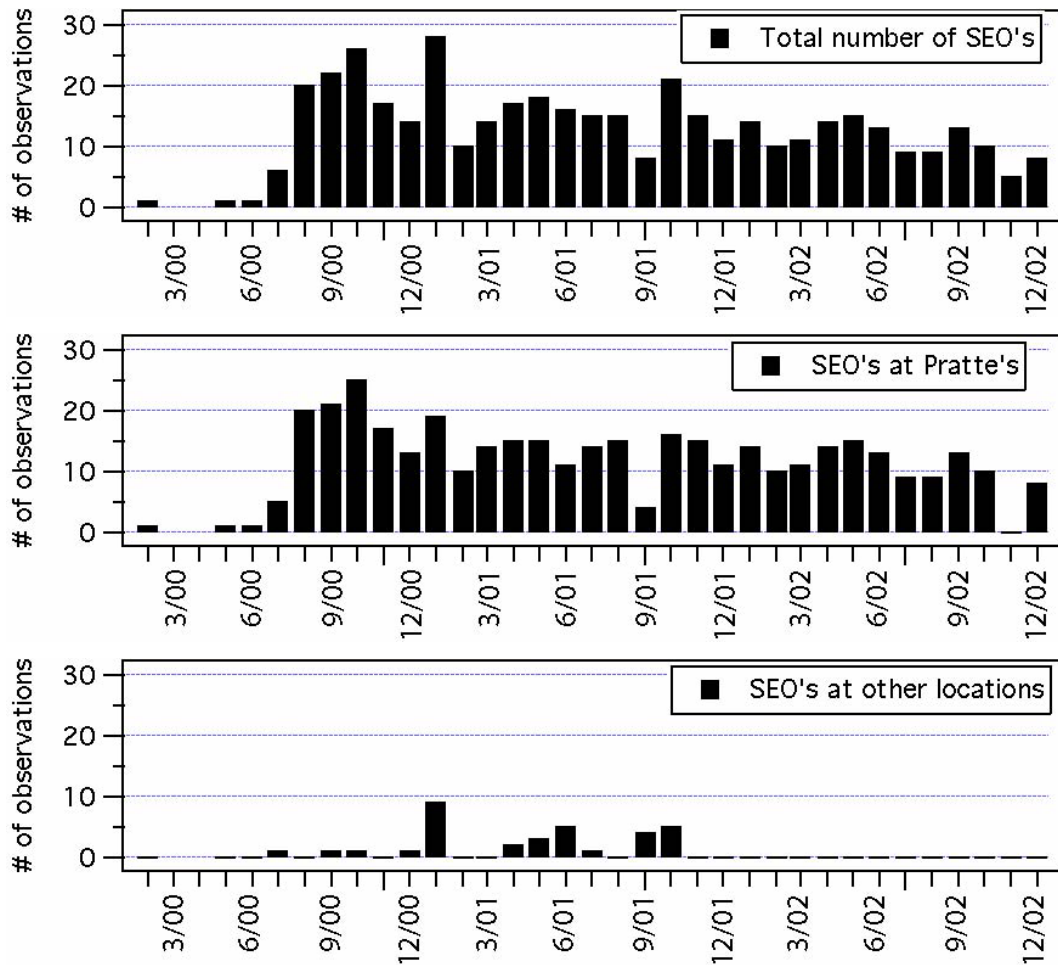


Figure 13: Breakdown of SEO observations by month.

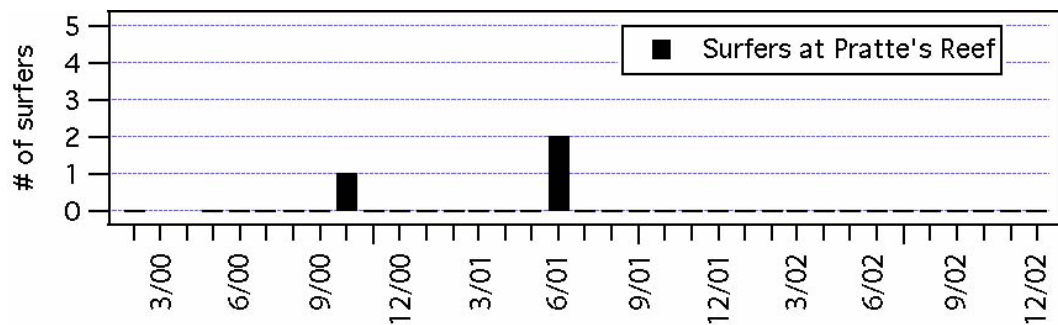


Figure 14: Breakdown of numbers of surfers present by month according to SEO data.

In addition, the SEO's recorded information on wave height, wave period and an approximation of the longshore current. Wave periods were estimated by having the observer count the number of seconds required for 11 wave crests to pass a certain point. Each observer estimated wave heights visually. This is a fairly subjective observation, however these data are relatively consistent since over 90% of the observations were recorded by the same individual. The longshore current was estimated by having the observer toss a floating drogue (generally a grapefruit, but sometimes an orange or a tennis ball) into the swash zone and recording the distance along shore that the drogue traveled before returned to land.

Figure 15 shows a comparison between the wave height and wave period as estimated by the SEO observer and the same data as recorded on the NOAA buoy #46025 located 30 miles west of the reef site. The observed values for each quantity were averaged for each month and compared to both the monthly average recorded on the buoy and a historical average recorded between 1982 and 1993. It is interesting to note that the observed wave periods match quite well with both sets of the recorded wave period data.

A large discrepancy is seen between the estimated and recorded wave heights. Not only are the observations consistently smaller, but they also appear to be out of phase with the monthly averages. One would expect for the observations to be smaller at this location. It is common knowledge that the waves along this stretch of beach are generally smaller than at El Porto, a well-known and popular surfing area to the south. El Porto is known for producing breakers larger than the buoy readings, while this part of the beach generally produces breaker heights smaller than what is being recorded offshore. A possible reason for the data being out of phase is that the SEO wave height data estimates the breaking wave height on the innermost sand bar. During periods of high surf (generally December – February) the waves are much larger on the outer bar and smaller on the inside. The larger outside waves however are not consistently recorded in the SEO, and therefore not reflected here.

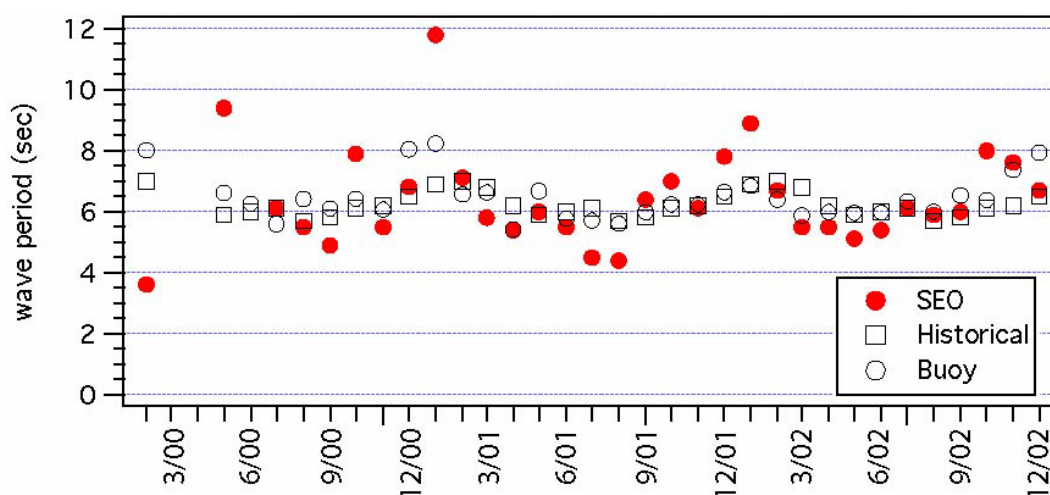


Figure 15: Comparison between observed and recorded wave period data. SEO is the data as estimated by the volunteers, historical is an average computed over the period 1982 – 1993, and Buoy is the average computed for that particular month.

Another quantity recorded by SEO observers is the distance and direction traveled by a drogue thrown into the near shore swash zone. The data collected are presented in Figure 17. It can be seen that there are slightly more south drift days over north drift. While this data is interesting, it only gives the direction and an indication of the strength of the current, it does not give any indication of an actual velocity since the time needed to travel that distance is not recorded.

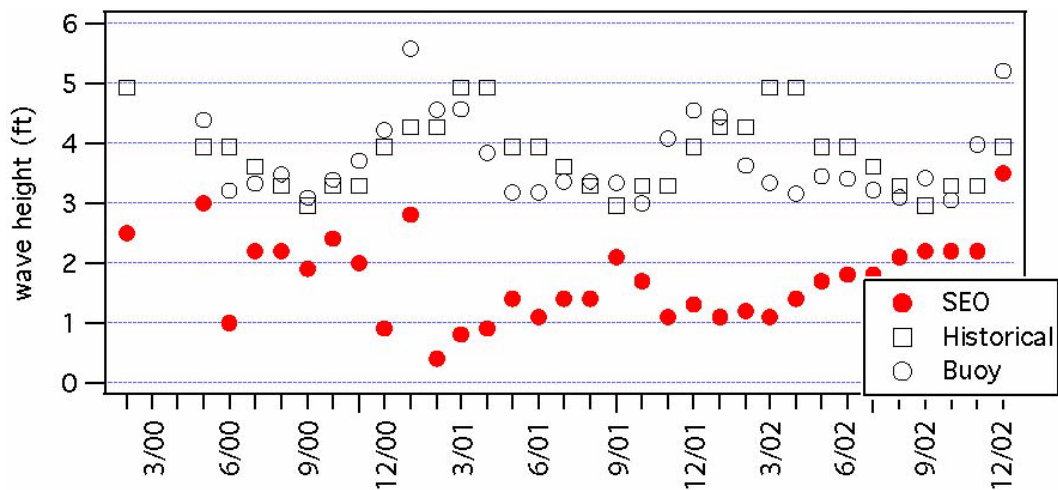


Figure 16: Comparison between observed and recorded wave height data.

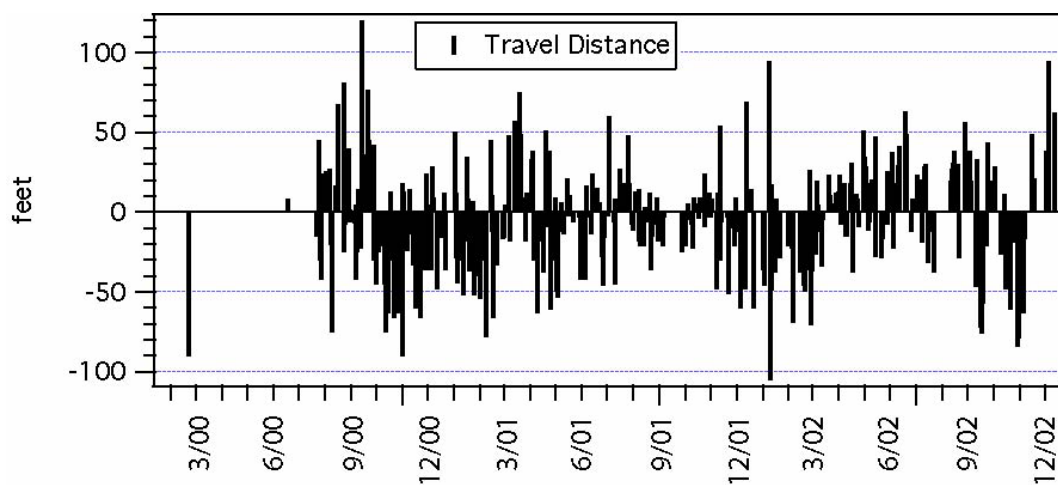


Figure 17: Drogue travel distances. Positive values are to the right (north).

Discussion: Surfing at Pratte's Reef

Over the two years of Pratte's Reef, there has been somewhere on the order of 10 surf sessions at the reef (most of those included the author). This is very disappointing when compared to the number of people that surfed a few km south at El Porto (tens to hundreds every day) or 200 meters north at Shitpipe (up to 10 at least bimonthly). However, the number of times Pratte's was surfed was probably the same or greater than any other spot between the El Segundo jetty and Shitpipe (not counting the jetty or Shitpipe itself). This demonstrates that Pratte's did create a minor improvement in the surfability of an otherwise unsurfed stretch of beach. However, this small number of surfers could also have been attracted by the novelty of the nation's first artificial surfing reef.

Almost all of the surfing that takes place between the El Segundo Jetty and Marina Del Rey is dependent on a man made structure to create surfable waves. This an important point to consider when discussing artificial reefs and surfing in central and southern Santa Monica Bay.

The surfing at Pratte's generally took place in the late winter or spring months (February through May). This is probably because at that time of year, the swells are generally peakier (swells with mixed direction) and of a shorter period, produced by local winds rather than distant storms. These

shorter swells are able to respond to the small dimensions of the reef while longer period swells passed over the reef unaffected. Figure 18 shows how the reef looked when it did break. It must be emphasized that Pratte's did produce rideable waves and improved the surfability of that section of the beach -- albeit temporarily.



Figure 18: Pratte's Reef breaking on November 12, 2001. Peakky westerly swell.



Figure 19: The same swell, the same day, 2 km south at El Porto.

Why Pratte's Reef didn't work as a surf spot.

In an academic sense, Pratte's reef was a success. It was a moral victory in that a corporate entity was held responsible for the destruction of a surfing area and it was a technical victory in that rideable waves were produced by a man-made structure. However, the reef did not attain the goal of mitigating for degraded surf. The simple proof is in the fact that basically no one surfed at the reef. Due to the density of surfers in the area, it is clear that if the reef had created good surfable waves, it would have been surfed.

The basic problem with Pratte's Reef is its small size, the volume and size of Pratte's Reef was limited by funding. Delays due to the resolution of permit issues for the El Segundo Groin

construction and permitting the reef itself compounded the problem by delaying the construction for many years. The maximum amount of construction material was purchased with the available funds; we now know that was not enough to significantly alter wave breaking and near shore coastal processes in the area. Figure 19 shows a comparison between Pratte's reef and the Narrowneck Reef in Australia. Narrowneck is 70 times bigger than Pratte's in terms of volume. Narrowneck extends over 150 m offshore, while Pratte's only extends 30. It is this cross-shore dimension that is the most important when it comes to manipulating incoming swells.

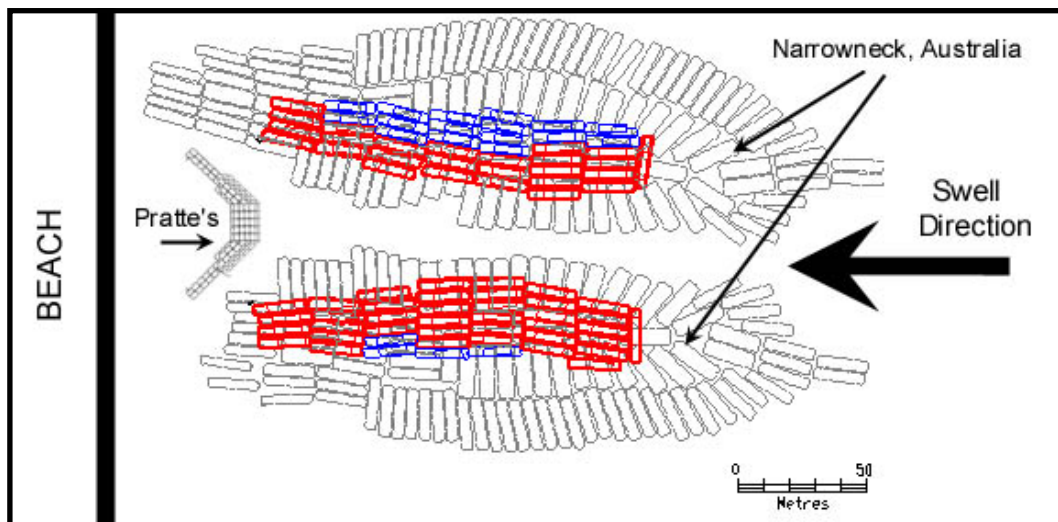


Figure 20: A comparison between Pratte's Reef and Narrowneck Reef.

Pratte's reef is in a good location for a surf enhancement structure. The beach fronting Pratte's is generally not good for surfing. It is very steep and reflective, there is a lot of "backwash", and incoming waves generally break simultaneously (close out), which makes them unsurfable. Good sandbars rarely form between the El Segundo Jetty and Marina Del Rey. The entire stretch of beach is under utilized and dominated by a series of shore perpendicular structures (jetties and sewer outfalls) that compartmentalize and stabilize the beach. The beach is wide and stable and there are no houses. There is nothing but the stable beach that could be harmed by the construction of an offshore structure. The swell direction is limited; this part of Santa Monica Bay receives swells only from the WNW to the WSW, between 300 and 240 degrees. Although the swell direction is limited, there is still a lot of swell that gets in to the area. In the months between October and April, the area receives its best swells from the west and northwest. In the winter and spring there are good surfing waves on at least a bi-weekly basis.

Along Dockweiler Beach the two most popular surfing locations are either created or enhanced by man made structures. Hammerland, the left break off of the north side of the El Segundo Jetty, can be a world-class wave under the right conditions. It is a very powerful, fast breaking, hollow wave that has been featured numerous times in surfing magazines. Furthermore, 200 meters north of Pratte's Reef is the 1-mile outfall from the Hyperion Sewage treatment plant (see Figure 2). At low tide the

pipe is exposed, however at high tide, waves break over the pipe in a consistent manner. Local surfers affectionately refer to this spot as ‘Shitpipe’ Figure 21 shows a series images taken of one wave at Shitpipe. Two riders share a peak and are able to complete several maneuvers each, this is the objective of any artificial surfing reef. Though unintentional, the very existence of a spot like Shitpipe demonstrates that an artificial surfing reef could work in this area. It also shows that the underperformance of Pratte’s reef as a surf spot can be attributed to deficiencies in the design rather than problems with the location or a lack of suitable swell. Future endeavors into artificial surfing reefs in Santa Monica Bay should look closely at the Hyperion 1-mile Outfall as a starting point for the design.



Figure 21: Splitting a peak Shitpipe, Los Angeles’ other “artificial reef”.

Conclusion

An artificial surfing reef known as “Pratte’s Reef” was constructed in El Segundo, west of Los Angeles, California. Constructed of approximately 1600 cubic meters of sand contained in 200 geotextile bags, the reef was designed to modify the wave breaking characteristics of incoming swells into a form more compatible with recreational surfing. A comprehensive monitoring program was implemented immediately after construction to definitively assess any impacts, positive or negative, that Pratte’s Reef might have on shoreline morphology, offshore bathymetry, marine biology and recreational surf quality. The results of this monitoring program show that Pratte’s Reef had essentially no impact on the shoreline or offshore morphology. Biological activity was seen to increase around the reef, as several species of marine animals used the reef for shelter. Pratte’s Reef did improve local surfing conditions, but only temporarily. Observations showed that the first installation of reef bags in September 2000 had only minimal, if any, impact on the wave breaking characteristics. The second installation in April 2001 did create much better surfing waves, but only for a few months and under the right conditions. The shallow crest created during the April 2001 installation was eroded away over the following winter by large swells that broke beyond the reef on outer sand bars. At present Pratte’s Reef is no longer affecting incoming swell waves in any way. The reef bags are mostly level with the sand. The underperformance of Pratte’s Reef as a surf spot can be largely attributed to deficiencies in the design, rather than other factors such as a poor location or a deficient wave climate. This is apparent since two popular surfing locations, both attributed to man made structures, lie on either side of Pratte’s reef – Shitpipe at the Hyperion Sewer Outfall 200 m to the north and Hammerland at the El Segundo Jetty 1 km to the south. The legacy of Pratte’s Reef should be the California Coastal Commission’s precedent that recreational surfing areas are State resources that deserve protection and cannot be destroyed as a result of coastal development.